

BRYAM STEVEN LÓPEZ MIRANDA

Software Engineer - Applied AI, Integrations and Voice AI

San Carlos, Alajuela, Costa Rica | Open to remote work in Latin America

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Professional summary

Computer Engineer specializing in applied AI, systems and product engineering. I build tool-using agents, RAG, memory, deterministic guards, automation, data pipelines, voice communications and user-facing web products. I deliver complete products with TypeScript, Next.js, Node.js, PostgreSQL, Supabase, Python, n8n, Asterisk, SIP and Retell, from the data model through to running them on Linux.

Professional experience

Systems Integration Engineer

AI Solutions CR | Mar 2025 - Present

- Designed and developed a multi-tenant conversational AI SaaS for WhatsApp, Messenger, Instagram, web and telephony.
- Built Plica Licitaciones CR, a procurement-intelligence engine over ten years of Costa Rican public spend (SICOP, Jul 2016-Jul 2026): 269k tenders, 349k awards and over one million comparable lines, with legal RAG on pgvector and a 24-tool MCP server.
- Built assistants with memory, RAG, tools, guards, fallbacks and per-organization configuration; one brain routes five brands within a single tenant.
- Integrated Voice AI with Asterisk 20, PJSIP, AMI, SIP trunks, Retell, Telnyx and DIDWW for inbound calls, outbound campaigns and transfers.
- Implemented RLS, feature flags, idempotency, auditing, outbox, notifications and failure recovery on Linux, PostgreSQL and self-hosted Supabase.

Software Developer - Professional Internship

Vivo Gaming | Jul 2024 - Nov 2024

- Developed an observability framework for Node.js services, centralizing metrics, traces and diagnostics with OpenTelemetry, Prometheus, Grafana and Elasticsearch.
- Documented the architecture and adoption procedures; the work became my Computer Engineering graduation project.

Technical skills

- Applied AI: AI Agents, LLMs, RAG, tool use, MCP, prompt engineering, deterministic guards, Retell.
- Software engineering: TypeScript, JavaScript, Python, Next.js, React, Node.js, REST APIs, multi-tenant systems.
- Data: PostgreSQL, Supabase, pgvector, data pipelines, RLS, predictive models.
- Voice AI and telephony: Asterisk 20, PJSIP, AMI, SIP trunks, Kamailio, RTPengine, Retell, Telnyx, DIDWW, LiveKit.
- Automation and infrastructure: n8n, Linux, Docker, PM2, nginx, Vercel, OpenTelemetry, Grafana, Prometheus.

Selected projects

- Plica Licitaciones CR: public-procurement intelligence with verifiable SICOP data, specialists, cited legal answers and honest predictions. Stack: TypeScript, Next.js, PostgreSQL, Supabase, Python, RAG, MCP. plica.aisolutionscr.tech
- Voice AI / Voice Ops: inbound and outbound calls, campaigns, transfers, browser SIP console, recordings and operational analytics. Stack: Asterisk 20, PJSIP, AMI, SIP, Kamailio, Retell, LiveKit. ops.aisolutionscr.tech
- Retell Flowkit: local engine to generate, fix, audit and validate Retell Conversation Flows; 751/751 tests and 28/28 official templates verified. Stack: TypeScript, skills, LLMs, validators, testing, Codex.
- Multi-tenant conversational AI SaaS: chat and voice agents with organization isolation, memory, RAG, tools, guards, human handoff, auditing and failure recovery. Stack: Next.js, TypeScript, Node.js, PostgreSQL, Supabase, n8n. app.aisolutionscr.tech/es
- Ya Sale: QR menu and ordering with diner, table and kitchen connected; updatable menus and availability, kitchen statuses and service or bill requests. Built with TicaCode. Stack: TypeScript, Next.js, Product Engineering, QR ordering, real-time operations. yasale.app

Education

Bachelor's Degree in Computer Engineering — Costa Rica Institute of Technology (TEC), 2020 - 2024. Graduation project: observability framework for Node.js services.

Certifications

JavaScript Algorithms and Data Structures (freeCodeCamp); Mobile Application Development (Google Activate); Professional C++ Course (Azul School).

Languages

Spanish (native) · English (technical reading; basic conversation)